

Education

Bachelor of Science in Biology (Computational Biology)
Florida State University, Tallahassee, FL

Expected May 2027
3.677 GPA

Research Experience

Undergraduate Research Fellow

(June 2026 - August 2026)

Strand Lab, *University of Texas Southwestern SURF*, Dallas, Texas

- Built a single-cell and spatial atlas of the human seminal vesicle and vas deferens, combining scRNA-seq and snRNA-seq data
- Designed a Xenium spatial transcriptomics panel to validate cell-type assignments and recover tissue architecture lost during digestion
- Presented poster at symposium

CPRIT-CURE Intern

(June 2025 - August 2025)

Ken Chen Lab, *MD Anderson Cancer Center CPRIT CURE Program*, Houston, Texas

- Programmed Python and R scripts to extract molecular interactions from curated databases (DoRothEA, Cellphone, STRING, KEGG) and assembled them into a directed knowledge graph
- Applied personalized PageRank over the graph to prioritize gene-signaling events
- Presented poster at symposium

SMART Intern

(June 2024 - August 2024)

Shaulsky Lab, *Baylor College of Medicine SMART Program*, Houston, Texas

- Performed wet lab procedures including plasmid preparation, bacterial growth, electroporation, GoldenBraid cloning, and restriction enzyme digestion
- Maintained lab journal for documenting daily procedures and presentation at weekly lab meetings
- Orally presented at end of program

Undergraduate Research Assistant

(August 2023 - Present)

Lemmon Lab, *Florida State University UROP*, Tallahassee, Florida

- Generate synthetic data using MatLab
- Program simulations and recorded peaks in parameter space in MatLab and R
- Presented at symposium and expo

Research/Presentations

Ho I., Paterson-Rees E., Malewska A., Lafin J., & Strand D. (2026). *A Single and Spatial Atlas of the Human Seminal Vesicle and Vas Deferens*. [Poster Presentation] UTSW SURF Poster Symposium, Dallas, TX

Ho I., He S., Mohanty V., & Chen K. (2025). *Integrating molecular signature databases to holistically reveal gene signaling events*. [Poster Presentation] MD Anderson CPRIT CURE Poster Symposium, Houston, TX

Ho I., Kurasawa M., & Shaulsky G. (2024). *Exploring the expression of chromoproteins in Dictyostelium discoideum*. [Oral Presentation] Baylor College of Medicine SMART Program, Houston, TX

Weintraub C., Ho I., & Lemmon A. (2024). *Modeling Neural Circuits to Understand Reproductive Isolation and Cryptic Evolution*. [Poster Presentation] Florida State University Department of Scientific Computing Exposition (CX24), Tallahassee, FL

Ho I., & Lemmon A. (2024). *Modeling Neural Circuits to Understand Incipient Speciation Part 2: Quantifying the Potential for Cryptic Evolution*. [Poster Presentation] Florida State University Undergraduate Research Opportunity Program End of Year Poster Symposium, Tallahassee, FL

Honors/Awards

- UTSW SURF 2026
- MD Anderson Cancer Center CPRIT CURE 2025
- Baylor College of Medicine SMART 2024
- FSU Dean's List 2023-2025
- FSU Honors Program Member 2023-present
- FSU UROP (Undergraduate Research Opportunity Program) 2023-2024

Community Service

Teacher

- Read English books and gave English presentations to elementary school students in Taiwan (May 2025)

Flower Mound Key Club

- Served the Flower Mound/Lewisville community in different services (November 2020 - May 2023)
- *Keep Flower Mound Clean*: Cleaned in various areas around Flower Mound

Teaching Assistant

- Co-taught English with teachers in Taiwan to grade school students (May 2021 – June 2021)

Bluebonnet Classroom

- Taught English to international grade school students weekly via Zoom (October 2020 - July 2021)

Skills

Language: English, Chinese (Beginner spoken), Spanish (first level, Spoken)

Computational: MATLAB, R and RStudio, Python, Seurat | Operating systems: MacOS, Linux

Project management: GitHub

Computer Applications: Google office (Word processing, Spreadsheet, Slides)

Revised: July, 2026